

4.7 Change of Variables in Multiple Integrals; Jacobians

Motivation: Consider the integral: $\int_0^2 2x\sqrt{x^2+1}dx$. We know from calculus one since $f(x) = 2x\sqrt{x^2+1}$ is continuous on $[0, 2]$ then by using a u -substitution we know that if $u = x^2 + 1$, then $du = 2x dx$ and the bounds become $u(0) = (0)^2 + 1 = 1$ and $u(2) = (2)^2 + 1 = 5$. Then, $\int_0^2 2x\sqrt{x^2+1}dx = \int_1^5 \sqrt{u} du = \frac{2}{3} u^{3/2} \Big|_1^5 = \frac{2}{3} (5^{3/2} - 1)$.

In this process we see that a few things have changed in order to complete the substitution:

- (1) The bounds (i.e., the interval we are integrating over).
- (2) The function we are integrating.
- (3) The differential dx had to be changed by finding the relationship between dx & du .

In Calculus III we have been integrating functions of two variables and we wish to see how we might be able to perform a uv -substitution (what we more commonly call a *change of variables*). When doing this our substitution will require that we change similar features of our integral.

- (1) The region that we are integrating over.
- (2) The function we are integrating.
- (3) The differential dA_{xy} ($dx dy$ or $dy dx$) will need to be changed to dA_{uv} ($du dv$ or $dv du$) by finding the relationship between dA_{xy} & dA_{uv} .

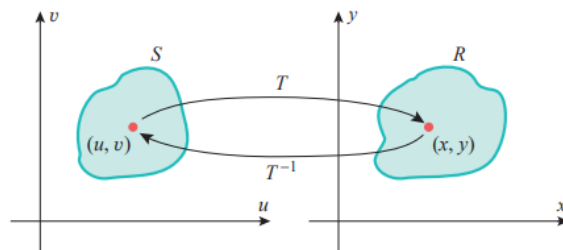
Before we look at how each of these changes work within a given uv -substitution we first introduce some important terminology for this section.

Definition: A transformation from the uv -plane to the xy -plane is a function of the form $T(u, v) = (g(u, v), h(u, v))$. Intuitively, we think of $g(u, v)$ as giving us an x value, and we think of $h(u, v)$ as giving us a y value.

Note: In function mapping notation we will write $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ since T takes in an ordered pair and spits out an ordered pair.

Definition: If $T(a, b) = (c, d)$ we say that (c, d) is the image of the point (a, b) under the transformation T . If no two points have the same image, T is called one-to-one.

Note: If T is one-to-one this implies that T has an inverse transformation we call T^{-1} which maps points from the xy -plane to the uv -plane.



Example 1: Consider the transformation given by $T(u, v) = (u + v, v^3)$.

a) Express the substitutions this transformation implies.

Here we have $x = u + v$ and $y = v^3$

b) What is the image of $(2, 3)$ under T ?

We calculate that: $T(2, 3) = (2 + 3, 3^3) = (5, 27)$.

c) Suppose that $(0, 0)$ is a point in the uv -plane, where does T map this point in xy -plane?

We calculate that: $T(0, 0) = (0 + 0, 0^3) = (0, 0)$.

d) Is T a one-to-one transformation?

Let's assume T is not one-to one and try to get a contradiction. Assume two different points (a, b) and (c, d) have the same image under T . This means that $b^3 = d^3 \Rightarrow b = d$. It also means $a + b = c + d \Rightarrow a + b = c + b \Rightarrow a = c$. This means that (a, b) and (c, d) must be the same point! This is a contradiction. So, T is one-to-one.

Now we turn our attention to how uv -substitutions affect the region we are integrating over.

Example 2: Consider the transformation given by $T(u, v) = (u^2 - v^2, 2uv)$. Find the image of the following square in the uv -plane: $S = \{(u, v) | 0 \leq u \leq 1, 0 \leq v \leq 1\}$ under the transformation T .

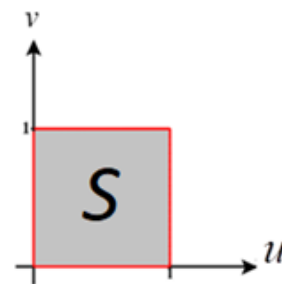
We will analyze all 4 boundary curves which make up S , and we will look at their image under T .

For $u = 0$ where $0 \leq v \leq 1$, we have that under T : $x = -v^2$ and $y = 0$. So, the image is: $y = 0$ where $-1 \leq x \leq 0$.

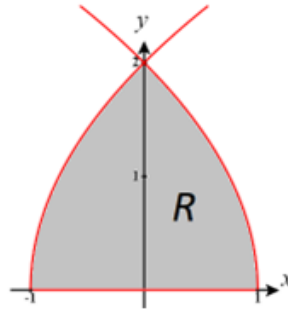
For $u = 1$ where $0 \leq v \leq 1$, we have that under T : $x = 1 - v^2$ and $y = 2v$. So, the image is: $x = 1 - \left(\frac{y}{2}\right)^2$ where $0 \leq y \leq 2$.

For $v = 0$ where $0 \leq u \leq 1$, we have that under T : $x = u^2$ and $y = 0$. So, the image is: $y = 0$ where $0 \leq x \leq 1$

For $v = 1$ where $0 \leq u \leq 1$, we have that under T : $x = u^2 - 1$ and $y = 2u$. So, the image is: $x = \left(\frac{y}{2}\right)^2 - 1$ where $0 \leq y \leq 2$.



A picture of the new region R is shown below:



Fact: If in the transformation $T(u,v) = (f(u,v), g(u,v))$ both $f(u,v)$ and $g(u,v)$ are linear functions in u and v then the transformation T maps lines to lines OR lines to points. The same holds for T^{-1} .

Example 3: Suppose we are going to work with a substitution where $u = 3x - 2y$ & $v = x + y$ (note this is a T^{-1} transformation). If this substitution is used on a region R in the xy -plane, shown below, it will create a region S in the uv -plane. Determine the region S in the uv -plane.

We see that the transformation here is composed of linear functions and so we know that linear boundaries of R will map to linear boundaries of S .

So, we see where the transformation takes corner point of R .

We see that under this transformation

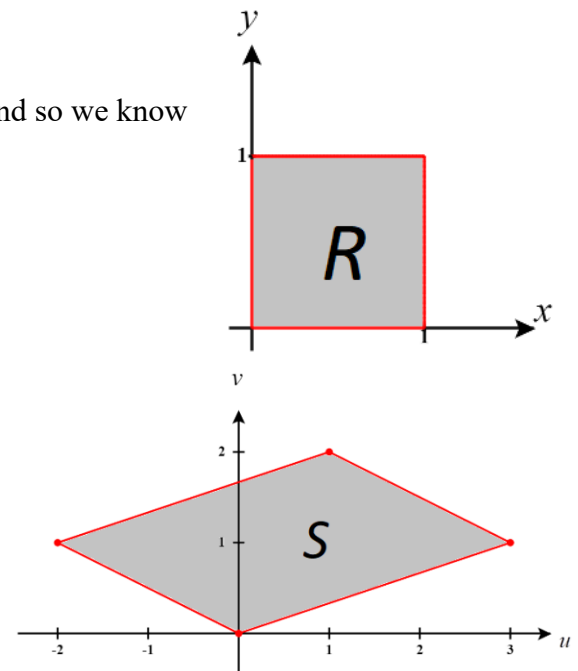
$(0,0)$ maps to $(0,0)$

$(1,0)$ maps to $(3,1)$

$(1,1)$ maps to $(1,2)$

$(0,1)$ maps to $(-2,1)$

So we see the region S shown in the graph at the right.



The past several examples have shown us how given transformations affect the region of integration.

Now the question is: How does our new region relate to the differentials that we start with and want to switch to and what does this imply about the areas of the regions that we have?

Example 4: Suppose that we are integrating $\iint_R 1 dA$ where R is a closed region with boundary given by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

- a) Show how the substitution given by the transformation $T(u,v) = (3u, 2v)$ gives us an easier region to integrate over.

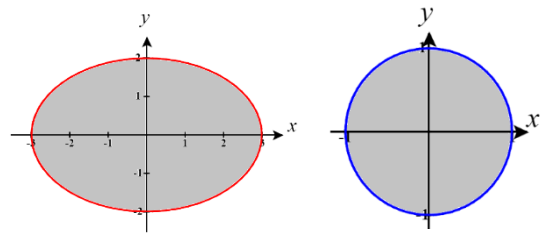
We see that the given region R is region shown on the left (with a red border).

The given transformation implies that $x = 3u$ and $y = 2v$. If we use this substitution we see that

$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \text{ becomes } u^2 + v^2 = 1 \text{ which is a circle of radius 1.}$$

The new region S , which is obtained after the substitution, is shown on the right (with a blue border).

Noticed that integrating over the region S will be easy to do since we could switch to using polar coordinates.



- b) Determine the relationship between dA_{xy} & dA_{uv} base on this substitution.

We see that if $x = 3u$ & $y = 2v$ then $dx = 3du$ & $dy = 2dv$ This implies that $dx dy = 6 du dv$

- c) Compute $\iint_R 1 dA$ by using the given uv -substitution.

$$\iint_R 1 dA_{xy} = \iint_R 1 dx dy = \iint_S 6 du dv = 6 \iint_S du dv = 6 (\text{Area of circle w / radius 1}) = 6 (\pi (1)^2) = 6\pi$$

When we established the relationship between $du dv$ & $dx dy$ we can think of the 6 we found as a “scaling factor” of the areas. So, the equation $dx dy = 6 du dv$ tells us that the area of the ellipse is 6 times larger than the area of the circle created by the substitution.

Note: It is not always the case that the area of R will scale uniformly. The next example shows this.

Example 5: Suppose we are going to work with a substitution where $u = \frac{x^3}{3}$ & $v = \frac{y^2}{2}$ over a region R . Determine the relationship between the area R and its transformation S .

We see that if $u = \frac{x^3}{3}$ & $v = \frac{y^2}{2}$ then $du = x^2 dx$ & $dv = y dy$.

This implies that $dudv = x^2 y dx dy \rightarrow dx dy = \frac{1}{x^2 y} dudv$

This shows us that the area of the region R scales at different rates for different points (x, y) .

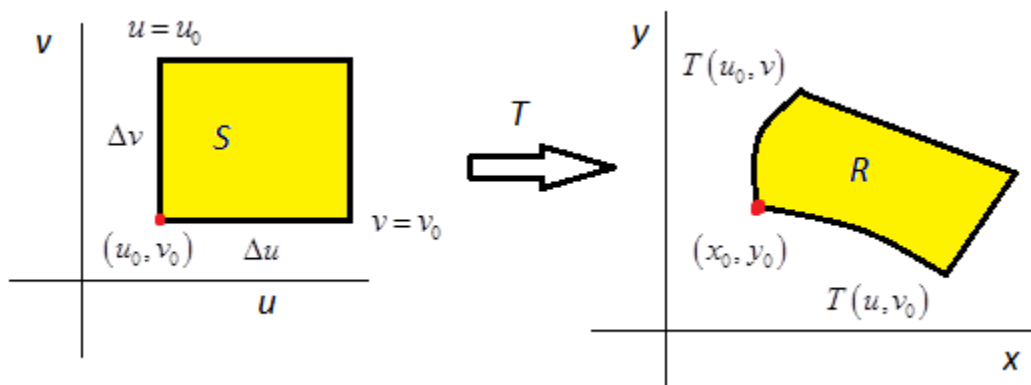
Note: It is also possible that finding such a relationship between $dudv$ & $dx dy$ is not so simple as in the previous two examples. For example, consider if we worked with a substitution like $u = 3x - 2y$ & $v = x + y$.

This leads us to ask: Is there a generalized way of determining this “scaling factor”?

Exploration (Not done in class):

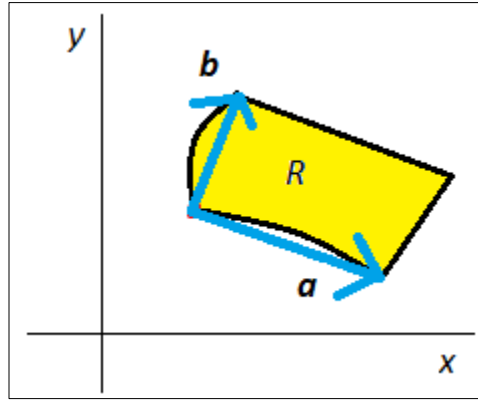
Suppose that $T(u, v) = (x(u, v), y(u, v))$ is a transformation with continuous first-order partial derivatives.

Suppose that S is a rectangle in the uv -plane (within the domain of T), and the side lengths of S are Δu and Δv . Consider the image, R , of S in the xy -plane under T . To get an idea of how a transformation of variables affects a double integral, we will try to approximate the area of R . A picture of the situation is:



Using the notation of the picture, we suppose that $T(u_0, v_0) = (x_0, y_0)$. We can think of T in the form of a parameterization. Specifically, let $\mathbf{r}(u, v) = \langle x(u, v), y(u, v) \rangle$. Then, a parameterization for the boundary of R given by $T(u, v_0)$ is $\mathbf{r}(u, v_0)$, and a parameterization for the boundary of R given by $T(u_0, v)$ is $\mathbf{r}(u_0, v)$.

Similar to section 4.4, we will approximate the area of R with a parallelogram. Specifically, we introduce vectors $\mathbf{a}$ and $\mathbf{b}$ as pictured below:



We notice that the initial point of both vector $\mathbf{a}$ and $\mathbf{b}$ is the terminal point of the position vector: $\mathbf{r}(u_0, v_0)$.

Also, the terminal point of vector $\mathbf{b}$ is the terminal point of the position vector: $\mathbf{r}(u_0, v_0 + \Delta v)$, and the terminal point of vector $\mathbf{a}$ is the terminal point of the position vector: $\mathbf{r}(u_0 + \Delta u, v_0)$. So, we have that

$$\mathbf{b} = \mathbf{r}(u_0, v_0 + \Delta v) - \mathbf{r}(u_0, v_0) \text{ and } \mathbf{a} = \mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0).$$

$$\text{We notice that } \mathbf{r}_u = \lim_{\Delta u \rightarrow 0} \frac{\mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0)}{\Delta u} \text{ and } \mathbf{r}_v = \lim_{\Delta v \rightarrow 0} \frac{\mathbf{r}(u_0, v_0 + \Delta v) - \mathbf{r}(u_0, v_0)}{\Delta v}.$$

$$\text{So, } \mathbf{a} = \mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0) \approx \Delta u \mathbf{r}_u(u_0, v_0) \text{ and } \mathbf{b} = \mathbf{r}(u_0, v_0 + \Delta v) - \mathbf{r}(u_0, v_0) \approx \Delta v \mathbf{r}_v(u_0, v_0).$$

This means that the area of the region R is approximately:

$$\|\Delta u \mathbf{r}_u(u_0, v_0) \times \Delta v \mathbf{r}_v(u_0, v_0)\| = \|\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)\| \Delta u \Delta v \text{ where we let the third component of } \mathbf{r} \text{ equal 0.}$$

Specifically, the approximate area of R is:

$$\|\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)\| \Delta u \Delta v = \left\| \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & 0 \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & 0 \end{vmatrix} \right\| \Delta u \Delta v = \left\| \left\langle 0, 0, \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \right\rangle \right\| \Delta u \Delta v = \left\| \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \right\| \Delta u \Delta v.$$

Where the derivatives above should be evaluated at (u_0, v_0) .

Definition: The Jacobian of the transformation T given by $T(u, v) = (x(u, v), y(u, v))$ is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \text{ (assuming these partial derivatives exist).}$$

Note: It can be useful to think of the Jacobian as consisting of the gradient of the x function in its first row and the gradient of the y function in its second row.

Note: Intuitively we can think of the Jacobian as an “area correction/scaling factor”. As it turns out the Jacobian appears when we make a change of variables in a double integral.

Example 6: Find the Jacobian of the transformation given by: $T(r, \theta) = (r \cos(\theta), r \sin(\theta))$.

We calculate that $\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{vmatrix} = r \cos^2(\theta) + r \sin^2(\theta) = r$

Notice that since r is defined as a nonnegative value this implies $dx dy = r dr d\theta$.

Note that this implies that $\iint_R f(x, y) dA = \iint_S f(r \cos(\theta), r \sin(\theta)) r dr d\theta$!

Note: We just derived the formula that we derived in section 4.3 via Riemann sums!

Theorem (Change of Variables in a Double Integral):

Suppose that T is a transformation with continuous first-order partials, whose Jacobian is nonzero and that T maps a region S in the uv -plane onto a region R in the xy -plane. Suppose that f is continuous on R and that R and S are type I or type II plane regions. Suppose that T is one-to-one, except perhaps on the boundary of S . Then,

$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

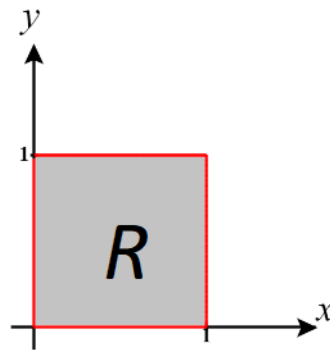
Note: The precise statement of conditions under which this formula works is beyond the scope of this course. The formula However, the theorem intuitively makes sense based upon our exploration above.

Note: We can also apply the above theorem when R and S can be decomposed into type 1 and type 2 regions (though this may require the use of more than one double integral). Also note that the order of the differentials on the right side of the equation may be taken to be $dv du$.

Example 7: Suppose that $R = \{(x, y) | 0 \leq x \leq 1 \text{ \& } 0 \leq y \leq 1\}$. Compute $\iint_R x^2 y dA$ by changing $u = x$ & $v = xy$.

First, we note that our function of integration $f(x, y) = x^2 y$ is everywhere continuous and that a picture of the region R is shown at the right.

We will find the region S in the uv -plane corresponding to R . Note that since the transformation is NOT composed of only linear function we cannot just map our corner points. Instead, we will consider what the transformation does to each edge of R .



For $x = 0$ where $0 \leq y \leq 1$, we have that under T : $u = 0$ and $v = 0$. So, the image is just a point $(0, 0)$

For $x = 1$ where $0 \leq y \leq 1$, we have that under T : $u = 1$ and $v = y$. So, the image is: $u = 1$ where $0 \leq v \leq 1$.

For $y = 0$ where $0 \leq x \leq 1$, we have that under T : $u = x$ and $v = 0$. So, the image is: $v = 0$ where $0 \leq x \leq 1$.

For $y = 1$ where $0 \leq x \leq 1$, we have that under T : $u = x$ and $v = x$. So, the image is: $v = x$ where $0 \leq x \leq 1$.

So we see the region S shown in the graph at the right.

We describe S as $S = \{(u, v) | 0 \leq v \leq u \text{ \& } 0 \leq u \leq 1\}$

Now we will find the Jacobian of our transformation in order to find how the area of S relates to the area of R .

To find the Jacobian we need to solve our equations for x and y .

Since $u = x$ \& $v = xy$ then $x = u$ \& $y = \frac{v}{u}$

Our transformation T for our change of variables is: $T(u, v) = \left(u, \frac{v}{u}\right)$.

We note that T has continuous first-order partials, and its Jacobian is:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -\frac{v}{u^2} & \frac{1}{u} \end{vmatrix} = \frac{1}{u}. \quad \text{This means that } dx dy = \left|\frac{1}{u}\right| du dv.$$

Note the Jacobian of T is defined as long as $u \neq 0$ (assuming $u \neq 0$ will not affect the integral).

We now use the change of variables theorem, and we obtain:

$$\begin{aligned} \iint_R x^2 y \, dA &= \iint_S uv \left|\frac{1}{u}\right| du dv = \int_0^1 \int_0^u v dv du = \int_0^1 \left(\frac{1}{2} v^2\right) \Big|_0^u du \\ &= \int_0^1 \frac{1}{2} u^2 du = \left(\frac{1}{6} u^3\right) \Big|_0^1 = \frac{1}{6} - 0 \\ &= \frac{1}{6} \end{aligned}$$

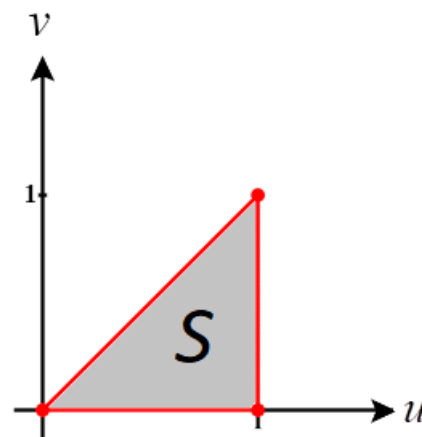
Example 8: In the last example we had to solve for x and y in order to find the Jacobian to complete the change

of variables. Compute $\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$. How does this compare to the Jacobian?

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ y & x \end{vmatrix} = x$$

Note that based on the substitution $\frac{\partial(u, v)}{\partial(x, y)} = x = u$. This means that this is the inverse of the Jacobian! This

implies that $du dv = |x| dx dy$



Remark: The product $\frac{\partial(x,y)}{\partial(u,v)} \cdot \frac{\partial(u,v)}{\partial(x,y)} = 1$. This means that the Jacobian and $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ are inverses of one another (assuming these partial derivatives exist).

Proof (not done in class):

We know that by definition $\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$ and that $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$.

Therefore,

$$\frac{\partial(x,y)}{\partial(u,v)} \cdot \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} \cdot \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = (x_u y_v - x_v y_u)(u_x v_y - u_y v_x) = x_u y_v u_x v_y - x_u y_v u_y v_x - x_v y_u u_x v_y + x_v y_u u_y v_x$$

We note however that

$$\begin{aligned} \begin{vmatrix} x_u u_x + x_v v_x & x_u u_y + x_v v_y \\ y_u u_x + y_v v_x & y_u u_y + y_v v_y \end{vmatrix} &= (x_u u_x + x_v v_x)(y_u u_y + y_v v_y) - (x_u u_y + x_v v_y)(y_u u_x + y_v v_x) \\ &= x_u u_x y_u u_y + x_u u_x y_v v_y + x_v v_x y_u u_y + x_v v_x y_v v_y - (x_u u_y y_u u_x + x_u u_y y_v v_x + x_v v_y y_u u_x + x_v v_y y_v v_x) \\ &= x_u u_x y_v v_y + x_v v_x y_u u_y - x_u u_y y_v v_x - x_v v_y y_u u_x \end{aligned}$$

Note that this implies that
$$\frac{\partial(x,y)}{\partial(u,v)} \cdot \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} \cdot \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = \begin{vmatrix} x_u u_x + x_v v_x & x_u u_y + x_v v_y \\ y_u u_x + y_v v_x & y_u u_y + y_v v_y \end{vmatrix}.$$

Now, by the chain rule we have that

$$\frac{\partial(x,y)}{\partial(u,v)} \cdot \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} \cdot \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = \begin{vmatrix} x_u u_x + x_v v_x & x_u u_y + x_v v_y \\ y_u u_x + y_v v_x & y_u u_y + y_v v_y \end{vmatrix} = \begin{vmatrix} x_x & x_y \\ y_x & y_y \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

■

Remark: If $dxdy = |J| du dv$ then $\frac{1}{|J|} dxdy = du dv$

Proof (follows from the above remark).

Note that these recent facts imply that to determine the scale factor between the regions R and S we can always just compute J or $\frac{1}{J}$, whichever is easiest!

We can now give a rough outline of the steps we take when performing a change of variables in a double integral.

Steps for Performing a Change of Variables in a Double Integral:

1. Let $x = x(u, v)$ and $y = y(u, v)$ OR $u = u(x, y)$ and $v = v(x, y)$ (sometimes we will have to determine these formulas, and sometimes these formulas will be given).
2. Draw a picture of the region R of integration and give inequalities that determine the region R
3. Ensure that the Transformation T (or T^{-1}) determined by the equations in 1 is a transformation with continuous first order partials and calculate the Jacobian (or the inverse Jacobian).
4. Determine the region S in the uv -plane. It may be helpful to solve the equations in 1 for x and y first.
5. Verify the hypotheses of the change of variable theorem are met (i.e. the Jacobian of T is nonzero, the function of integration is continuous on R , and T is one-to-one except perhaps on the boundary of S).
6. Apply the change of variables formula:
$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

Note: In MTH 202 we will not always write out all the details for step 5, but we will at least mentally verify the hypotheses.

Example 9: Consider the double integral $\iint_R y dA$ where $R = \left\{ (x, y) \left| \frac{y^2}{4} - 1 \leq x \leq 1 - \frac{y^2}{4}, y \geq 0 \right. \right\}$. Use the change of variables $x = u^2 - v^2$ $y = 2uv$ to find this double integral.

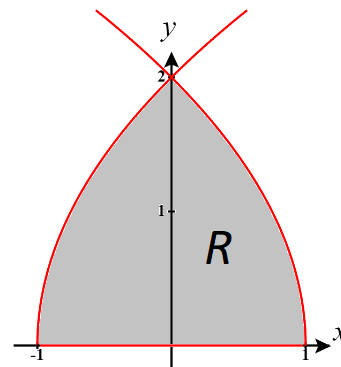
First, we note that our function of integration $f(x, y) = y$ is everywhere continuous and that a picture of the region R is shown at the right.

Our transformation T for our change of variables is: $T(u, v) = (u^2 - v^2, 2uv)$.

We note that T has continuous first-order partials.

Based on the substitution given we compute the Jacobian:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix} = 4u^2 + 4v^2.$$



Note the Jacobian of T is nonzero unless $(u, v) = (0, 0)$ (assuming $(u, v) \neq (0, 0)$ will not affect the integral).

Notice that we already found the region S in the uv -plane corresponding to R in a previous example. We know that $S = \{(u, v) | 0 \leq u \leq 1, 0 \leq v \leq 1\}$

We now use the change of variables theorem (note the hypotheses are satisfied), and we obtain:

$$\begin{aligned} \iint_R y dA &= \iint_S 2uv |4u^2 + 4v^2| du dv = \int_0^1 \int_0^1 (8u^3v + 8uv^3) du dv = \int_0^1 \left(2u^4v + 4u^2v^3 \right) \Big|_{u=0}^{u=1} dv \\ &= \int_0^1 (2v + 4v^3) dv = \left(v^2 + v^4 \right) \Big|_0^1 = 2. \end{aligned}$$

Note: In the past problems we have been given a uv -substitution to use, but in general this is not always the case. Much like in Calculus I we must find our own substitutions to use. Our main goals when doing this are:

- (1) Make sure our integrand is easy to work with (we have seen examples of how integrands setup one way may be more difficult than in others).*
- (2) Make sure that the region that we are integrating over is easy to work with (we have seen examples of moving to easier regions in polar, spherical, and cylindrical coordinates).*

Example 10: Consider the integral $\iint_R e^{(x+y)/(x-y)} dA$ where R is the trapezoidal region with vertices $(1,0)$, $(2,0)$, $(0,-2)$, & $(0,-1)$.

a) What is a possible change of variables that will make this integral easier to evaluate.

Since finding an antiderivative of $e^{(x+y)/(x-y)}$ with respect to x or y would be difficult, we get the idea to let $u = x + y$ and $v = x - y$.

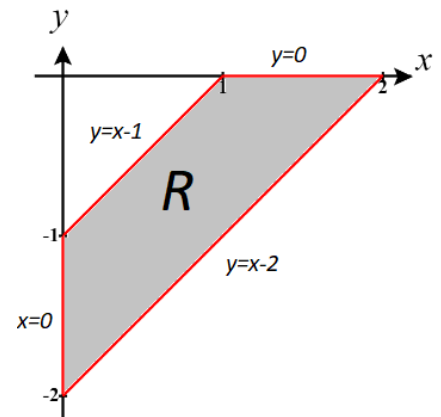
b) Use the change of variables from (a) to evaluate the integral.

A picture of the region R of integration is shown at the right.

We have that $T^{-1}(x, y) = (x + y, x - y)$ is a transformation with first-order

partials and the inverse Jacobian is $J^{-1} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -1 - 1 = -2$ Therefore,

$$dudv = |-2| dx dy \rightarrow dx dy = \frac{1}{2} dudv$$



Now, we find S in the uv -plane whose image is R under T . Note that since the transformation is composed of only linear function we could just map our corner points, however we will show work for mapping each edge for practice.

For $y = x - 2$ where $0 \leq x \leq 2$ we have that $u = x + (x - 2) = 2x - 2$ and $v = x - (x - 2) = 2$.

This gives $v = 2$ where $-2 \leq u \leq 2$.

For $y = x - 1$ where $0 \leq x \leq 1$ we have that $u = x + (x - 1) = 2x - 1$ and $v = x - (x - 1) = 1$.

This gives $v = 1$ where $-1 \leq u \leq 1$.

For $y = 0$ where $1 \leq x \leq 2$ we have that $u = x + (0) = x$ and

$$v = x - (0) = x.$$

This gives $v = u$ where $1 \leq u \leq 2$.

For $x = 0$ where $-2 \leq y \leq -1$ we have that $u = 0 + y = y$ and

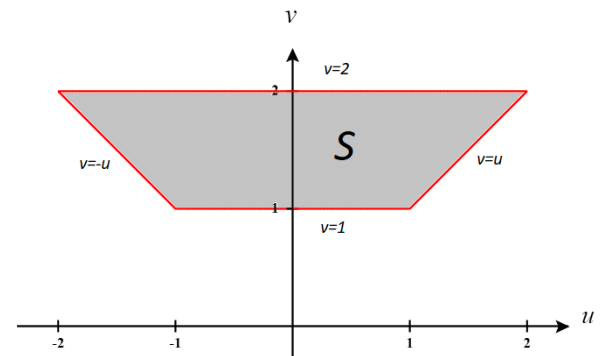
$$v = 0 - y = -y.$$

This gives $v = -u$ where $-2 \leq u \leq -1$.

So, the region S is: $S = \{(u, v) | -v \leq u \leq v, 1 \leq v \leq 2\}$.

So, we have

$$\iint_R e^{(x+y)/(x-y)} dA = \iint_S e^{u/v} \frac{1}{2} dudv = \frac{1}{2} \int_1^2 \int_{-v}^v e^{u/v} dudv = \frac{1}{2} \int_1^2 v e^{u/v} \Big|_{u=-v}^{u=v} dv = \frac{1}{2} \int_1^2 v (e - e^{-1}) dv = \frac{e - e^{-1}}{2} \left(\frac{v^2}{2} \right) \Big|_1^2 = \frac{3(e - e^{-1})}{4}.$$



We end 4.7 by briefly discussing how the change of variables theorem generalizes to triple integrals.

Definition: Suppose that T is a transformation from uvw -space to xyz -space given by $T(u, v, w) = (g(u, v, w), h(u, v, w), k(u, v, w))$. If the first order partials of the component functions of T are continuous, then the Jacobian of T is the three-by-three determinant:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

Fact: Under similar hypotheses to the change of variables for a double integral theorem, it is possible to prove a theorem for the change of variables of a triple integral. The formula for the change of variables is:

$$\iiint_B f(x, y, z) dV = \iiint_S f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw.$$

Example 11: Suppose that E is the region in 3-space given in spherical by

$$E = \{(\rho, \theta, \phi) : \alpha \leq \theta \leq \beta, c \leq \phi \leq d, g_1(\theta, \phi) \leq \rho \leq g_2(\theta, \phi)\}.$$

Then, E is the image of the region $S = \{(\rho, \theta, \phi) : \alpha \leq \theta \leq \beta, c \leq \phi \leq d, g_1(\theta, \phi) \leq \rho \leq g_2(\theta, \phi)\}$ in the $\rho\theta\phi$ -space under the transformation $T(\rho, \theta, \phi) = (\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi))$.

Use the change of variables fact just mentioned to find a formula for $\iiint_E f(x, y, z) dV$ (you may assume f is continuous on E).

The Jacobian of the transformation T is:

$$\begin{aligned} \frac{\partial(x, y, z)}{\partial(\rho, \theta, \phi)} &= \begin{vmatrix} \sin(\phi) \cos(\theta) & -\rho \sin(\phi) \sin(\theta) & \rho \cos(\phi) \cos(\theta) \\ \sin(\phi) \sin(\theta) & \rho \sin(\phi) \cos(\theta) & \rho \cos(\phi) \sin(\theta) \\ \cos(\phi) & 0 & -\rho \sin(\phi) \end{vmatrix} \\ &= \sin(\phi) \cos(\theta) [-\rho^2 \sin^2(\phi) \cos(\theta) - 0] - (-\rho \sin(\phi) \sin(\theta)) [-\rho \sin^2(\phi) \sin(\theta) - \rho \cos^2(\phi) \sin(\theta)] \\ &\quad + \rho \cos(\phi) \cos(\theta) [0 - \rho \sin(\phi) \cos(\theta) \cos(\phi)] \\ &= -\rho^2 \sin^3(\phi) \cos^2(\theta) - \rho \sin(\phi) \sin(\theta) [\rho \sin(\theta)] - \rho^2 \sin(\phi) \cos^2(\theta) \cos^2(\phi) \\ &= -\rho^2 \sin^3(\phi) \cos^2(\theta) - \rho^2 \sin(\phi) \sin^2(\theta) - \rho^2 \sin(\phi) \cos^2(\theta) \cos^2(\phi) \\ &= -\rho^2 \sin(\phi) [\sin^2(\phi) \cos^2(\theta) + \sin^2(\theta) + \cos^2(\theta) \cos^2(\phi)] = -\rho^2 \sin(\phi) [\cos^2(\theta) + \sin^2(\theta)] \\ &= -\rho^2 \sin(\phi). \end{aligned}$$

Since we know that $0 \leq \phi \leq \pi$, we have $\left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| = \left| -\rho^2 \sin(\phi) \right| = \rho^2 \sin(\phi)$.

So, $\iiint_E f(x, y, z) dV = \iiint_S f(\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)) \rho^2 \sin(\phi) d\rho d\theta d\phi$.

Note: This is the theorem we use for spherical coordinates from section 4.6.2!